FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Advanced)-2019

PAPER –1

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01– 06, 19 – 24, 37 - 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*

   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 - 46) *contains 12 questions. Based on this section contains* ***TWO (02)*** *paragraphs. Based on each paragraph, there are* ***TWO (02)*** *questions.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A

(One or More than one correct type)

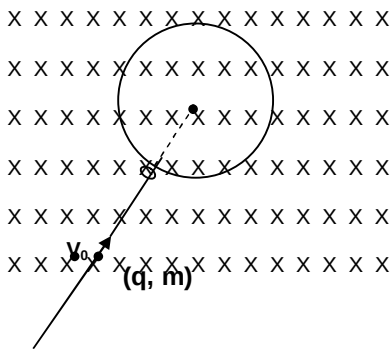
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. If λ_{min} is minimum wavelength produced in $X-$ray tube and $\lambda_{K\alpha}$ is the wavelength of K_α line. As the operating voltage is increased

(A) $\lambda_{K\alpha}$ increases (B) $\lambda_{K\alpha}$ decreases

(C) $(\lambda_{K\alpha} - \lambda_{min})$ increases (D) $(\lambda_{K\alpha} - \lambda_{min})$ decreases

2. A non-conducting cylindrical shell of radius R is placed in uniform magnetic field. The axis of the cylinder is parallel to the direction of magnetic field. A hole is drilled in cylinder and a particle having charge q and mass m is projected with velocity v_0 perpendicular to magnetic field and directed towards the axis of cylinder as shown in diagram. The particle collides elastically with inner wall of the cylinder and rebounds.



(A) The minimum no. of collisions after which it will emerge is two.

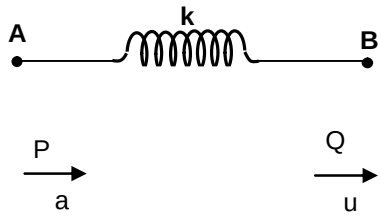
(B) The maximum speed v_0 for which it can come out is $v_0 = \left(\frac{qBR}{m}\right)\sqrt{3}$.

(C) The minimum time in which it can come out will be $\left(\frac{\pi m}{qB}\right)$

(D) If the particle is projected with a velocity $v_0/3$ $\left\{\text{where } v_0 = \left(\frac{qBR}{m}\right)\sqrt{3}\right\}$ it will come out after five collisions.

Space for Rough work

3. A massless spring of spring constant k has two ends P and Q as shown in the figure. End Q is being pulled by person B with constant speed u towards right. End P is being pushed by person A with constant acceleration a towards right starting from rest. Both starts simultaneously. Choose the correct option at the instant when B appears to be at rest with respect to A



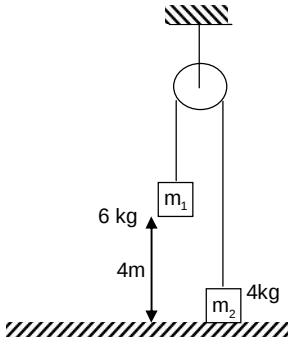
(A) work done by $B = ku^4 / 3a^2$

(B) work done by $A = -5ku^4 / 24a^2$

(C) work done by $A = +5ku^4 / 24a^2$

(D) elastic potential energy stored in spring $= ku^4 / 8a^2$

4. Two objects are connected by a light string passing over a light frictionless pulley as shown. The object of mass 6 kg is released from rest. ($g = 10\ m/s^2$)



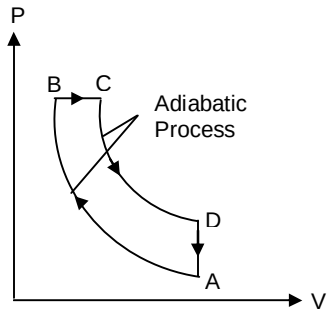
(A) Time taken by object of mass 6 kg to strike the surface is 2sec.

(B) Maximum height attained by block of mass 4 kg is 4.8 m.

(C) Duration after which 6 kg block will leave the surface for the first time is 2.8 sec.

(D) None of the above

5. An idealized diesel engine operates in a cycle known as air standard diesel cycle as shown. Fuel is sprayed into the cylinder at the point of maximum compression, B. Combustion occurs during expansion BC.



(A) Heat is absorbed during process BC

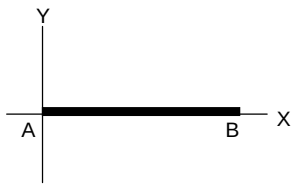
(B) Heat is released during process DA

(C) Heat is absorbed during process CD

(D) Efficiency of the cycle is $\left[1-\frac{1}{\gamma}\left(\frac{T_D - T_A}{T_C - T_B}\right)\right]$

Space for Rough work

6. A wire of mass M and length L is placed along x-axis with end A at origin. The linear mass density λ (of the rod) varies as $\lambda = \lambda_0 x$, where x is distance from end A, λ_0 is a constant.



(A) The value of λ_0 is $\frac{2M}{L^2}$

(B) Moment of Inertia of the rod about an axis passing through A and parallel to y-axis is $\frac{1}{2}ML^2$

(C) Moment of inertia of the rod about an axis passing through COM of the rod and parallel to y-axis is $\frac{ML^2}{18}$.

(D) Moment of inertia of the rod about an axis passing through B and parallel to y-axis is $\frac{1}{6}ML^2$.

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 7 and 8

A ring of radius R is made of a thin wire of material of density p, having cross-section area a and Young's Modulus y. The ring rotates about an axis perpendicular to its plane and through its centre. Angular frequency of rotation is ω.

7. The tension in the ring will be

(A) $\frac{apR^2\omega^2}{2}$ (B) $apR^2\omega^2$

(C) $2apR^2\omega^2$ (D) $\frac{apR^2\omega^2}{4}$

8. The ratio of kinetic energy to potential energy is

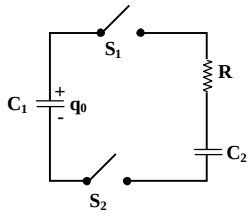
(A) $\frac{y}{pR^2\omega^2}$ (B) $\frac{2y}{pR^2\omega^2}$

(C) $\frac{y}{2pR^2\omega^2}$ (D) $\frac{y}{4pR^2\omega^2}$

Space for Rough work

Paragraph for Question Nos. 9 and 10

The capacitor C_1 in the figure shown initially carries a charge q_0. When the switches S_1 and S_2 are closed, capacitor C_1 is connected in series to a resistor R and a second capacitor C_2, which is initially uncharged. Where $C = \frac{C_1C_2}{C_1 + C_2}$.



9. The charge flown through wire as a function of time t is

(A) $q_0e^{-t/RC} + \frac{C}{C_2}q_0$ (B) $\frac{q_0C}{C_1} \times \left[1 - e^{-t/RC}\right]$

(C) $q_0\frac{C}{C_1}e^{-t/CR}$ (D) $q_0e^{-t/CR}$

10. The total heat dissipated in the circuit during the discharging process of C_1 is

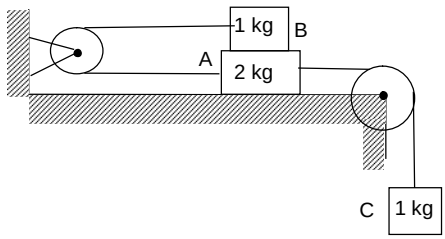
(A) $\frac{q_0^2}{2C_1^2} \times C$ (B) $\frac{q_0^2}{2C}$

(C) $\frac{q_0^2C_2}{2C_1^2}$ (D) $\frac{q_0^2}{2C_1C_2}$

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. Friction coefficient between any two surfaces in contact is 0.5. Pulleys and strings are frictionless. Find the friction between blocks A and B (in Newton)

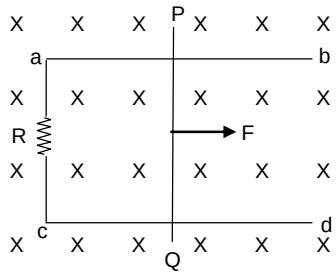


Space for Rough work

12. When a tangential force of 0.02 N is applied on a large wooden plate of area $10\,m^2$ floating on the surface of river, plate moves with the speed 2 m/s on the river surface. The river is 1 m deep and the water in contact with the bed is stationary. Assuming constant speed gradient, coefficient of viscosity of water is 10^{-N} poise. Find the value of N

13. A car is moving towards an approaching bike. Bike is moving with velocity 30 m/s. Frequency of horn sounded by car is 100 Hz while it is observed by bike rider as 120 Hz. The actual duration of horn is 6 sec. Find the time the bike rider hears it, in seconds. Take velocity of sound in air as 330 m/s.

14. In the figure ab and cd are two long conducting wires kept parallel to each other at a separation l in a uniform time varying magnetic field. The ends a and c are connected together by a resistor of resistance R. The magnetic induction B perpendicular to the plane of the wires varies with time according to the relation $B = B_0 t$, where B_0 is a positive constant with proper unit and t is the time in second. A conducting wire PQ placed on the wires ab and cd and dragged on the wire with constant speed v along the length of the wires by applying a constant force F. At $t = 0$, PQ is very close to ac. Ignore any resistance (electrical as well as mechanical) other than R. Find the value of $\frac{FR(250)}{B_0^2 l^2 v t^2}$.

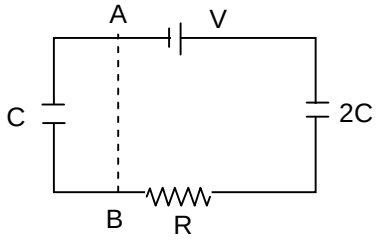


15. A choke coil is needed to operate an arc lamp at 250 V (rms) and 50 Hz. The lamp has an effective resistance of 15Ω when running at 10 A (rms). The inductance of the choke coil is $\frac{N}{125\pi}$H. Find the value of N

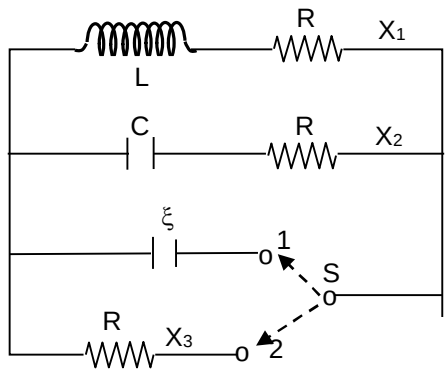
16. A conducting rod of length 1 m aligned along the unit vector $\left(\frac{3}{5}\hat{j}+\frac{4}{5}\hat{k}\right)$, is moving with a velocity $\vec{v} = (2\,\hat{i})\,m/s$. The motion take place in a uniform magnetic field $\vec{B} = (\hat{j} - 2\hat{k})$ tesla. Find the magnitude of induced emf in across the rod in volt.

Space for Rough work

17. If the heat dissipated in the circuit after a zero resistance wire is connected across the pairs A & B, is $\frac{2CV^2}{n}$, the value of 150n is



18. Figure shows a circuit containing three resistor X_1, X_2 & X_3 having resistance R each, an inductor, capacitor and an emf source having inductance 'L', capacitance C & emf ξ respectively (Given $R = \sqrt{\frac{L}{C}}$). The switch is first connected to position-1. When charge on capacitor becomes half of it maximum possible value then switch S is connected to position-2. Current in resistance X_3 just after shifting the switch from position-1 to position -2, is (given L = 5mH, C = $2\mu F, R = 10\Omega, \xi = 5V$)



Space for Rough work

Chemistry

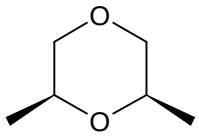
PART – II

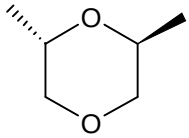
SECTION – A

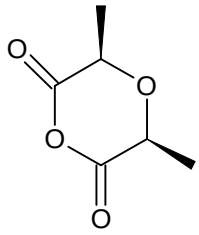
(One or More than one correct type)

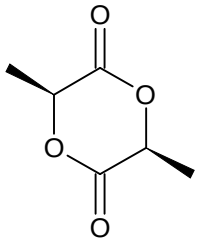
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

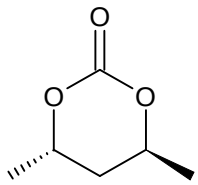
19. Identify which of the structures below are not meso structures?

1 

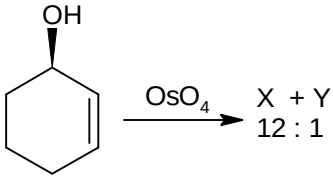
2 

3 

4 

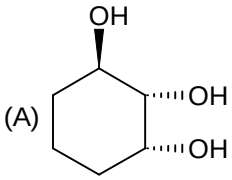
5 

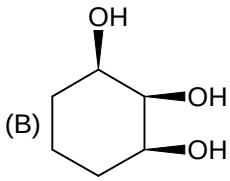
(A) 1 and 3
(B) 1, 3 and 5
(C) 1, 3 and 4
(D) 2 and 5

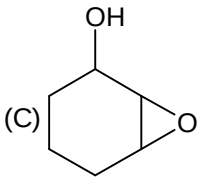
20. 

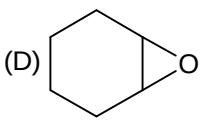
The product which is not X

Prodcut (X) will be :

(A) 

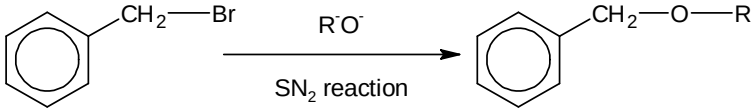
(B) 

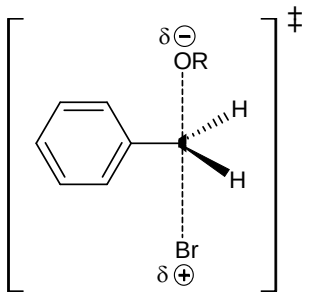
(C) 

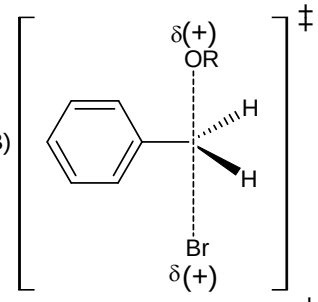
(D) 

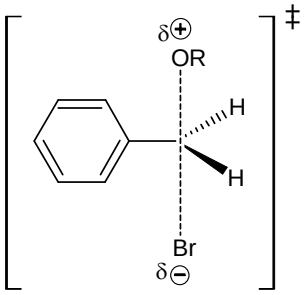
Space for Rough work

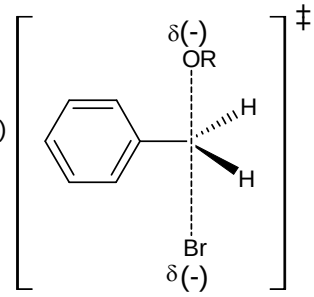
21. Which transition state is not for SN^2?



(A)


(B)


(C)


(D)


22. The correct statement(s) among the following is (are):
(A) All d-orbitals except d_{z^2} have two angular nodes
(B) $d_{x^2-y^2}$ and d_{z^2} orbitals lie on the axis or axes
(C) The degeneracy of p-orbital remains unaffected in a homogeneous external magnetic field
(D) The angle between the Z – component of orbital angular momentum of P_z – electron and Z-axis is 90°

Space for Rough work

23. The standard redox potential E^o of the following systems are

	System	E^o (volts)
(i)	$MnO_4^- + 8H^+ + 5e - \longrightarrow Mn^{2+} + 4H_2O$	1.51
(ii)	$Sn^{2+} \longrightarrow Sn^{4+} + 2e^-$	-0.15
(iii)	$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^3 + +7H_2O$	1.33
(iv)	$Ce^{3+} \longrightarrow Ce^{4+} + e^-$	-1.61

The oxidizing power of the various species are related as

(A) $Cr_2O_7^{2-} > MnO_4^-$ (B) $Ce^{4+} > Sn^{4+}$

(C) $Ce^{4+} > MnO_4^-$ (D) $MnO_4^- > Sn^{4+}$

24. Which of the following condition are correct for a binary solution, in which the solute as well as the solvent are liquid?

(A) C_6H_6 and $C_6H_5CH_3 \Delta H_{solution} > 0$; $\Delta V_{solution} = 0$

(B) CH_3COCH_3 and $CHCl_3 \Delta H_{solution} < 0$; $\Delta V_{solution} < 0$

(C) H_2O and $HCl \Delta H_{solution} > 0$; $\Delta V_{solution} < 0$

(D) H_2O and C_2H_5OH $\Delta H_{solution} > 0$; $\Delta V_{solution} > 0$.

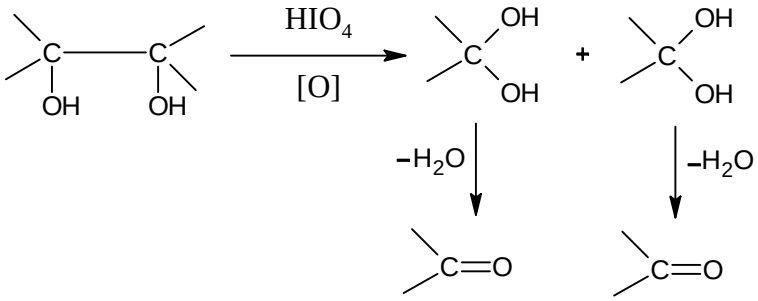
Space for Rough work

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

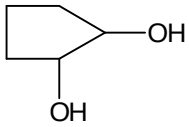
Paragraph for Question Nos. 25 and 26

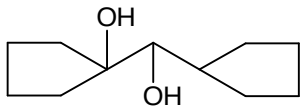
Vicinal diol on periodic acid oxidation gives a mixture of two carbonyl compounds which may be same of different.

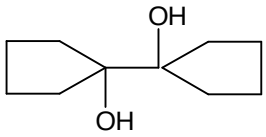


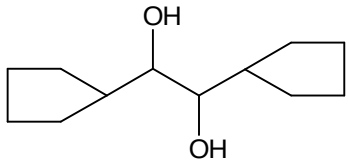
If diol is not vicinal then it is not oxidised by HIO_4. Cyclic diol on oxidation gives dicarbonyl compounds. By knowing the structures of carbonyl compounds formed we can get the structures of diols also.

25. 2 moles of cyclopentanone is obtained on HIO_4 oxidation of

(A) 

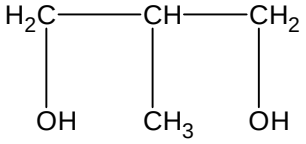
(B) 

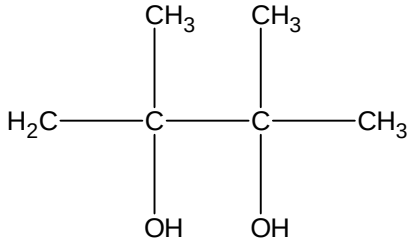
(C) 

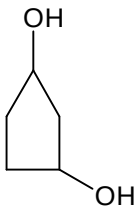
(D) 

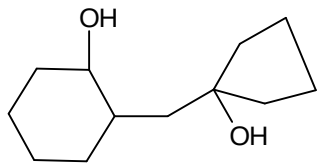
Space for Rough work

26. Which one of the following will be oxidised by periodic acid?

(A) 

(B) 

(C) 

(D) 

Paragraph for Question Nos. 27 and 28

Given

$E^{o}_{Ni^{2+}/Ni} = -0.26$ V

$E^{o}_{Sn^{2+}/Sn} = -0.12$ V

A solution containing $SnSO_4$(1M) and $NiSO_4$(1 M) is electrolyzed. (assume neutral aqueous solution)

27. Which of the ions will be deposited preferentially at cathode? (Given pH of pure water = 7) $\left(P_{H_2} = 1\,\text{atm}\right)$

(A) Ni^{2+} (B) Sn^{2+}

(C) H^{+} (D) OH^{-}

28. The concentration of Sn^{2+} when both Ni^{2+} and Sn^{2+} starts depositing at cathode. $\left(\text{assume } \frac{2.303RT}{F} = 0.06\right)$

(A) 1 M (B) 10^{-4}M

(C) 10^{-3}M (D) 10^{-8}M

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. Find the quantum number 'n' corresponding to the excited state of He^+ ion. If on transition to the ground state that ion emits two photons in succession with wavelengths 108.5 and 30.4 nm.

30. How many isomers of $C_4H_{10}O$ reacts with CH_3MgBr to evolve CH_4 gas? (Excluding stereoisomer)

31. Write the sum of geometrical isomers in $\left[Pt\left(H_2N-CH(CH_3)-CO\overset{\ominus}{O}\right)_2\right]$ complex and stereomers of $\left[Pt(gly)_3\right]^{\oplus}$ complex.

32. 2.47 g of an element A_x when dissolved in 50 g of benzene, lowered the freezing point of benzene to 0.4 K. K_f for benzene is 5.12 K kg mol^{-1}. The value of x, for atomic weight of A being 79, is

33. What is the time in hours for collecting 18 g of magnesium from molten $MgCl_2$ using a current of 10 amperes?

34. What $[H_3O^+]$ must be maintained in a saturated H_2S solution to precipitate Pb^{2+}, but not Zn^{2+} from a solution in which each ion is present at a concentration of 0.01 M?
(K_{sp} of $H_2S = 8\times10^{-22}$; K_{sp} of $ZnS = 1.0\times10^{-24}$)

35. A thermally isolated vessel contains 100 g of water at $0\ ^oC$. When air above the water is pumped out, some of the water freezes and some evaporates at $0\ ^oC$ itself. Calculate the mass of the ice formed such that no water is left in the vessel. (Latent heat of vaporization of water at $0\ ^oC = 2.10\times10^6$ J / Kg and latent heat of fusion of ice $= 3.36\times10^5$ J / Kg.)

36. A mineral having formula $Mg_xAl_yO_z$ is found in the spinel structure in which O^{2-} ions constitute ccp lattice, Mg^{2+} ions occupy $1/8^{th}$ of the tetrahedral voids and Al^{3+} ions occupy ½ of the octahedral voids. Find the total positive charge contained in one unit cell.

Space for Rough work

Mathematics

PART – III

SECTION – A

(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. $\int \frac{x^{2018}}{1+x+\frac{x^2}{2!}+.....+\frac{x^{2018}}{2018!}}\,dx = m!\,x - m!\log P(x) + c$, then

(A) $m = 2019$ (B) $m = 2018$

(C) $P(x) = 1 = x + \frac{x^2}{2!} + ... + \frac{x^{2018}}{2018!}$ (D) $P(x) = x + \frac{x^2}{2!} + ... + \frac{x^{m+1}}{(m+1)!}$

38. For $|x| > 1$, $\lim\limits_{n\to\infty} \prod\limits_{K=0}^{n} \left(1 + \frac{2}{x^{2^K} + x^{-2^K}}\right) = f(x)$, then

(A) $\int_2^5 f(x)\,dx = 3 + \ln 16$ (B) $\lim\limits_{x\to\infty} f(x) = 1$

(C) $f(x) = 0$ has one solution (D) $f(x)$ is decreasing function

39. If $f : R \to R$ be a differentiable function such that $(f(x))^7 = x - f(x)$, then

(A) The value of $\int_0^{\sqrt{2}} f^{-1}(x)\,dx$ is 3

(B) $f(x)$ is increasing for $\forall\ x \in R$

(C) The area bounded by by curve $y = f(x)$, between the ordinates $x = 0, x = \sqrt{3}$ & x axis is $\frac{f(\sqrt{3})}{8}\left[8\sqrt{3} - \left(f\sqrt{3})\right)^7 - 4\,f\left(\sqrt{3}\right)\right]$

(D) The area bounded by curve $y = f(x)$ between the ordinates, $x = 0, x = \sqrt{3}$ & x – axis is $f\left(\sqrt{3}\right)\left[\sqrt{3} + \left(f(\sqrt{3}\right)^4 + 4\,f\left(\sqrt{3}\right)\right]$

Space for Rough work

40. Let C_1 & C_2 be centres of two circles whose radii are 2 & 4 respectively. Also $C_1C_2 = 10$ & direct common tangents of these circles touch them at P, Q, R, S. Another circle of radius 'λ' is drawn passing through P, Q, R, S. Then

(A) Midpoint of $C_1\ C_2$ is centre of the circle passing through P, Q, R, S.

(B) Centre of the circle passing through P, Q, R, S divides $C_1\ C_2$ in the ration 1 : 2

(C) $\lambda^2 = 33$

(D) $\lambda^2 = 35$

41. Let $S(n+1) = \sum_{r=0}^{n} \frac{(-1)^{n-r} r}{r+1} \cdot \frac{(n+1)!}{r!(n+1-r)!}$ then

(A) $S(8) = 1$

(B) $S(7) = \frac{3}{4}$

(C) $S(9) = \frac{4}{5}$

(D) $S(11) = \frac{5}{6}$

42. Let $A(t) = [aij]$ is a matrix of order 3×3 given by $aij = \begin{cases} 2\cos t, \text{ if } i = j \\ 1 \quad \text{if } |i-j| = 1 \\ 0, \quad \text{otherwise} \end{cases}$

Where $|A(t)|$ is determinant value of matrix A(t), then which of the following are correct?

(A) $\lim_{t \to 0} |A(t)|\,|A(4t)|$ is equal to 16

(B) Maximum value of $|A(t)|\,|A(3t)|$ is equal to 16

(C) $\int_0^{\pi} |A(t)|\,|A(4t)|\,dt$ is equal to 0

(D) $\left|A\left(\frac{\pi}{17}\right)\right| \left|A\left(\frac{4\pi}{17}\right)\right|$ is equal to 1

Space for Rough work

(Paragraph Type)

This section contains **TWO** paragraphs. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 43 and 44

Let $f(x)$ satisfy the functional equation $(f(x))^2 = 1 + x\, f(x+1)$ & the inequality $\frac{x+1}{2} \le f(x) \le 2(x+1)\ \forall\ x \ge 1$, then answer the following questions.

43. The function $f(x)$ defined from $[1, \infty) \to R$ is

(A) Into (B) Onto
(C) Many one (D) None of these

44. The domain the function $y = \cos^{-1}[f(x)]$ is

(A) ϕ (B) $[-1, 1]$
(C) $[-2, 1]$ (D) None of these

Paragraph for Question Nos. 45 and 46

Let $< a_n >$ & $< b_n >$ be the arithmetic sequences each with common difference 2 such that $a_1 < b_1$ & let $c_n = \sum_{k=1}^{n} a_n$, $d_n = \sum_{k=1}^{n} b_k$. Suppose that the point $A_n(a_n, c_n)$, $B_n(b_n, d_n)$ are all lying on the parabola C: $y = px^2 + qx + r$ where p, q, r constants.

45. The value of p equals

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$
(C) $\frac{1}{2}$ (D) 2

46. If $r = 0$, then the value of a_1 & b_1 are

(A) $\frac{1}{2}$ & 1 (B) 1 & $\frac{3}{2}$
(C) 0 & 2 (D) $\frac{1}{2}$ & 2

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. Let $f(x) = 30 - 2x - x^3$ then the number of positive integral values of x which satisfies $f\left(f\left(f(x)\right)\right) > f\left(f\left(-x\right)\right)$ is

48. Let ABCD be a tetrahedron such that the edges AB, AC & AD are mutually perpendicular let the area of triangles ABC, ACD & ADB be given by 1, 3 & $\sqrt{90}$ sq. units respectively, then the area of ΔBCD is ___ sq. units.

49. Let p & q are real numbers such that
$(\tan p - 1)^3 + 2019\,(\tan p - 1) = -1$ & $(1 - \cot q)^3 + 2019\,(1 - \cot q) = -1$
$\tan p \neq \cot q$, then number of possible values of r which satisfy the equation
$\tan p + \cot q + \sin r + \cos r = 3$ in $x \in [-2\pi, 2\pi]$ is

50. Two circles are passing through point A of the Δ ABC & one of the circle touches the side BC at B & other circle touches the side BC at C. If $BC = 5$ & $\angle A = 30^o$ & the product of radii of two circles is

51. The area of region in xy plane consisting of all points (p, q) such that quadratic equation $px^2 + 2(p + q - 7)x + 2q = 0$ has less than 2 real solutions is ……., where (π = 3.14).

52. If $\bar{p}, \bar{q}, \bar{r}$ are three vectors such that $|\bar{p}| = |\bar{q}| = 1$ & $\bar{r} = \bar{p} \times \bar{q}$, then maximum value of $\left|(\bar{p} + 2\bar{q} + 3\bar{r}).[(2\bar{p} + 3\bar{q} + \bar{r}) \times (3\bar{p} + \bar{q} + 2\bar{r})\right|$ is equal to

53. Four distinct integer are picked at random from $\{0, 1, 2, 3, 4, 5, 6\}$. If the probability that among those selected the second smallest is 3 is P, then 35 P is equal to ……

54. Given $M = \left[\frac{1}{1^{2/3}} + \frac{1}{2^{2/3}} + ... + \frac{1}{1000^{2/3}}\right]$, where [.] denotes G.I.F., then (M –20) is equal to

Space for Rough work